Applied Machine Learning

Course Project

CS/DS 3262

Spring 2026

Announcement

HW3 was released on April 3 and is due on April 10 at 11:59 PM

Start working on it as soon as possible!

Remember to drop by office hours if you get stuck for more than 30 minutes!

DS/MS student Sophia Chung will visit our class and share tips and insights on finding DS jobs

Final Project

- **Gain hands-on experience** with data preprocessing, model selection, and evaluation in a real-world machine learning problem

- **Work on a meaningful project** instead of just assessments, applying concepts from class to a practical scenario

Final Project

This is a group project, with up to 3 students per team

Timeline:

- **Submit Proposal by** April 17, 11:59 PM
- **Final Submission (Report & Code)**: April 28, 11:59 PM

Looking for a project partner but haven't found one yet? Sign up using the Google link **before April 10**, and we'll try to match you with someone! 😊
https://forms.gle/GJH6uhwRdEhUTeE3A

Projects you can work on

- **Your own research Projects or from Independent Studies**: Any project related to machine learning that **demonstrates depth** and meaningful exploration

- **Deal Outcome Prediction from SymTrain (Industry Project):**
  - Predict whether a deal will be successful based on the provided company data
  - Gain industry experience working on a real-world problem
  - You will get additional **5 points** if you choose this one!

- **Song Recommendation:**
  - Build a content-based song recommendation system based on song features such as artist, genre, year, etc

What should be Included in Proposal

- **Names and student IDs** of all group members (up to 3 students)
- **Selected project** and a brief description of your understanding of the problem
- **Proposed approaches:** Outline some initial ideas for solving the problem (these may evolve later)
- **Length:** Maximum one page
- **Submission:** Only one team member should submit on behalf of the group

Deal Outcome Prediction

Data Provided by Symtrain

Project Description

SymTrain provides AI-powered coaching solutions to companies

The goal of this project is to build a **predictive model to forecast deal success to** help the sales team focus on high-potential opportunities

UNITED AIRLINES

Our company recently hired several new agents. Could you please provide training to help them better understand the business and learn how to assist customers effectively?

Sure, no problem! Here's the AI-powered training platform to help the new hires get started!

symtrain™

SymTrain sold a training plan to United Airlines to train new agents — **this is considered a successful deal**

Data Description

- **Files Used:**
  - anonymized_hubspot_companies.csv
  - anonymized_hubspot_deals.csv

- **Mapping File:** mappings.json

- **Data Merge Logic:** industry_project_data_merge.ipynb

- **Data Dictionaries:** overview of key variables from the PDFs

Deals Dataset

anonymized_hubspot_deals.csv

This dataset includes detailed information about all deals, such as their current status ("Is Deal Closed?"), whether they were won or lost ("Is Closed Won"), and the current stage of each deal ("Deal Stage").

Detailed description of each column can be found at **data-dictionary-deals.pdf**

	Deal Description	Days to close	Is Deal Closed?	Is Closed Won	Deal Stage
0	40-50 employees	69.0	False	False	Opportunity
1	3,000 + Agents.	149.0	False	False	Partner Referrals
2	5-6k agents	149.0	False	False	Partner Referrals
3	Furthest Along - 300 agents	149.0	False	False	Partner Referrals
4	BPO through partnership with AmplifAI	120.0	False	False	Deep Dive. PSP Drafted
...	...	...	...	...	...

Training data should only contain closed deal!

Companies Dataset

anonymized_hubspot_companies.csv

This dataset contains anonymized company records from Hubspot CRM. All personally identifiable information (PII) and sensitive business information has been removed or masked. This data describes client and prospect companies, their attributes, and technology stacks.

```
companies.head()
```

df

	Parent Company	CCaaS	Annual Revenue	Target Account	Associated Contact	Number of Form Submissions	Total Agents	Web Technologies	Close Date	# of Agents Total	...	BPO Program	Primary Sub-Industry	Number of Sessions	WFM	C
0	NaN	NaN	1.000000e+09	NaN	Contact_ef780380	0.0	NaN	Route 53	NaN	NaN	...	NaN	NaN	0.0	NaN	
1	NaN	NaN	NaN	NaN	Contact_93373ba5	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	
2	NaN	NaN	5.000000e+07	NaN	Contact_2e8e0993	0.0	NaN	Youtube; App Nexus; Google Tag Manager; Facebo...	NaN	NaN	...	NaN	NaN	0.0	NaN	
3	NaN	Five9	5.000000e+07	NaN	Contact_635e44ed	0.0	NaN	Postmark; Facebook Advertiser; Salesforce; Goo...	NaN	NaN	...	NaN	NaN	0.0	NaN	
4	NaN	Five9	1.000000e+08	NaN	Contact_8d055096	0.0	NaN	Microsoft Office 365; Google Tag Manager; Reca...	NaN	NaN	...	NaN	NaN	0.0	NaN	

Merge Datasets

The deals dataset does not include any company-related information, and the company dataset does not contain any deal-specific details (e.g., whether a deal was successful).

mappings.json file contains the dictionaries that can be used to map any of the 2 files together based on the unique IDs

```
### read in the mapping file
import json

with open('mappings.json', 'r') as f:
    mappings = json.load(f)

### check the keys
print(f'mapping keys: {mappings.keys()}')
```

```
mapping keys: dict_keys(['CompanyToDeals', 'CompanyToTickets', 'ParentChildRelationships', 'TicketToDeal'])
```

- CompanyToDeals: contains the mappings from company ID to Deals ID
- CompanyToTickets: contains the mappings from company ID to Tickets ID
- TicketToDeal: contains the mappings from Ticket ID to Deal ID

Merge the Deals and Company Dataset

```python
# revert the company-deal mapping
deal_to_company = {}

for company_id, deal_list in mappings["CompanyToDeals"].items():
    for deal_id in deal_list:
        deal_to_company[deal_id] = company_id
deal_to_company
```

```
{'29547238775': '17771181752',
 '9794988112': '9364308545',
 '9817334075': '9366087515',
 '9817334090': '9366136383',
 '9817320072': '9366140751',
 '9817329956': '9366155050'
```

Deal ID Company ID

Merge the Deals and Company Dataset

Add a company ID column in the deals dataset for table merging

```
### record ID is the unique deal ID
### first convert it into string type to match the datatype in the dictionary
deals['Record ID'] = deals['Record ID'].astype(str)

#### create a company for company id based on the dictionary using mapping function

deals['Company ID'] = deals['Record ID'].map(deal_to_company)
```

map function is used to create the company ID that is corresponding to the deal ID in the dels dataset

Merge the deals dataset and the companies dataset

commpany id from the deals table: 'Company ID'

company id from the company table: 'Record ID'

```
### convert the integer type to string type for Record ID
companies['Record ID'] = companies['Record ID'].astype(str)
```

```
deals_merged = pd.merge(left = deals, right = companies, left_on ='Company ID'  ,right_on ='Record ID' , how = 'left')
```

pd.merge is used to merge the deals and companies dataset together

Binary Classification Task: Deal Outcome Prediction

- **Goal:** Predict whether a deal will close successfully
- **Target variable:** Is Deal Closed
- **Models (3 examples):**
  - Logistic Regression
  - Random Forest
  - Decision Tree

Data Preprocessing Steps

- Handling missing values (drop, impute, or categorical “Unknown”)
- Detecting & treating outliers
- Encoding categorical variables (Label Encoding, One-Hot, frequency encoding)
- Feature scaling
- Feature correlation heatmap
- …

Exploratory Data Analysis (EDA)

- Data distribution of key features (e.g., histogram)
- Relationships between deal outcome and key features (e.g., pairplot)
- Highlight any surprising patterns
- …

Evaluation & Comparison

Tune at least one hyperparameter using K-fold cross-validation and select the best model based on the F1 score

Report the best model's performance using Accuracy, F1, Precision, Recall, and AUC

Model	Accuracy	F1 Score	Precision	Recall	AUC
Logistic Regression					
KNN					
Random Forest					
AdaBoost					
…					

Example of results table

Insights & Future Work

Top predictive features (e.g., company size, deal amount, ticket volume)

Insights:

- Example: what company characteristics increase the likelihood of a successful deal?

What else can be done or should be included?

Deliverables (For all Projects except for Song Recommender System)

1. **Raw Code Files**
   - Include all scripts or notebooks used for data processing, modeling, and analysis.
   - Missing raw code submission will result in a **deduction of 20 points**
2. **Final Report**

   The report should be well written and include the following sections:
   - **Problem Statement**
   - **Data Description**
   - **Data Preprocessing Steps**
   - **Exploratory Data Analysis (EDA)**
   - **Classification Model Development**
     - Model descriptions and hyperparameter tuning
   - **Insights & Future Work**

Grading Rubric for Final Report

1. **Problem statement (10 pts)**
   Clearly define the problem and your machine learning objective.
2. **Data Description (10 pts)**
   Describe your data (source, variables, size, and limitations).
3. **Data Processing (15 pts)**
   Clean and prepare the data (handle missing values, encode variables, scale/select features).
4. **EDA (10 pts)**
   Use visualizations or statistical analysis to explore the data and gain insights.
5. **Model development (20 pts)**
   Build and justify **at least 3** appropriate models.
6. **Training & Tuning (10 pts)**
   Explain how you train models and tune hyperparameters using **K-fold Cross validation**.
7. **Evaluation & Comparison (15 pts)**
   Use proper metrics and compare model performance.
8. **Insights & Future Work (10 pts)**
   Interpret results and suggest improvements or next steps.
9. **Report Quality (10 pts)**
   Clear writing, good organization, and reproducible code.

Evaluation & Comparison for Classification Task

Tune at least one hyperparameter using K-fold cross-validation and select the best model based on the F1 score

Report the best model's performance using Accuracy, F1, Precision, Recall, and AUC

Model	Accuracy	F1 Score	Precision	Recall	AUC
Logistic Regression					
KNN					
Random Forest					
AdaBoost					
…					

Example of results table

Evaluation & Comparison for Regression Task

Tune at least one hyperparameter using K-fold cross-validation and select the best model based on validation Root Mean Squared Error (RMSE).

Report the best model's performance using MSE, RMSE, MAPE, and R2

Model	MSE	RMSE	MAPE	R2
Linear regression				
KNN				
Random Forest				
AdaBoost				
…				

Example of results table

Song Recommender System

When a user inputs a song name through the interactive input, recommend **top-K similar songs** based on song features (e.g., genre, year)

To measure the distance or similarity of any 2 songs:

- Euclidean distance (measures the absolute distance)
- Cosine similarity (measures the relative similarity)

Dataset used to Build Song Recommender System

Your hometown	What language do yo	Song name	Artist	Genre	Language of th	Song release year
City	English;Chinese	Fame is a gun	Addison Rae	Pop	English	2020
Suburban	English;Chinese	run for the hills	Tate McRae	Pop	English	2020
Suburban	English	Hard to Explain	The Strokes	Indie / Alternative	English	2000–2009
Suburban	English	The Nights	Avicii	Pop	English	2010–2019
Suburban	English	Rock that body	Black eyed peas	Electronic / Dance	English	2010–2019
City	English	Wild Corsicians	Raekwon	Hip Hop / Rap	English	2020
Suburban	English	Murder on the dancefloo	Sophie Ellis-bextor	Pop	English	2020
City	English	Hey Ya!	Outkast	Pop	English	2000–2009
Suburban	English;Chinese	Renegade	Styx	Rock	English	1960–1979
Suburban	English;Chinese	Reverie for another Sphe	Taishi	Indie / Alternative	No lyrics	2010–2019
Suburban	English	Mr. Jones	Counting Crows	Alternative Rock	English	1980–1999
Suburban	English	Paranoid Android	Radiohead	Rock	English	1980–1999
Suburban	English	Pink pony club	Chappell Roan	Pop	English	2020
City	English	Levitating	Dua Lipa	Pop	English	2020
Suburban	English	Human	The Killers	Electronic / Dance	English	2000–2009

Dataset: ‘Song recommendations.csv’

Questions to Think About

1. What should your system do if a song entered by the user is not found in the dataset?
   Propose a solution and implement it.
   (Hint: You may prompt the user to provide features such as genre, artist, year, or context.)

2. Explore whether user attributes (e.g., age, major, preferences) are associated with certain types of songs.
   Provide at least one insight based on your analysis.

Questions to Think About

3. Some songs have multiple genres, languages, or tags. How will you represent these features in your model?
   (Hint: Consider techniques such as multi-hot encoding.)

Multi-hot encoding represents a data point with multiple categories by assigning a 1 to each applicable category and 0 to the rest.

Genre
Pop
Pop
Indie / Alternative
Pop
Electronic / Dance

Multi hot encoding

Pop	Indie	Alternative	Electronic	Dance
1	0	0	0	0
1	0	0	0	0
0	1	1	0	0
1	0	0	0	0
0	0	0	1	1

Deliveries for Song Recommender System

1. **Raw Code Files**
   - Include all scripts or notebooks used for data processing, modeling, and analysis.
   - Missing raw code submission will result in a **deduction of 20 points**
2. **Final Report**

Final Report for Song Recommender System

1. **Problem Statement:** Describe the recommendation task and your approach.
2. **Data Description:** Describe the dataset, including key variables (song features and user attributes), data size, and any data issues.
3. **Data Preprocessing:** Explain how you cleaned the data and handled features (e.g., multiple genres/tags).
4. **EDA:** Provide basic insights (e.g., genre distribution, patterns, data issues).
5. **User Preference Analysis:** Analyze relationships between user attributes and song preferences using correlation or other appropriate methods.
6. **Recommendation Method:** Explain how your system recommends songs (e.g., similarity measure).
7. Describe how your system handles songs not found in the dataset.
8. **Example Results:** Show sample recommendations and briefly explain why they make sense.